Sungwon Yang

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Research Interests

Legged locomotion, reinforcement learning, sim-to-real transfer, and real-world fine-tuning and continual adaptation of learned controllers.

Education

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea
M.S., Division of Future Vehicle *Mar 2026 – Spring 2027 (expected)*
Advisor: Prof. David Hyunchul Shim, Unmanned Systems Research Group (USRG)

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea
B.S. in Computer Science, Concentration in Artificial Intelligence *Sep 2017 – Feb 2026*
Leave of absence for mandatory military service, Oct 2023 – Apr 2025

Research Experience

Graduate Researcher, USRG, KAIST Mar 2026 – Present
Advisor: Prof. David Hyunchul Shim
Sim-to-real reinforcement learning pipeline for the Unitree Go2 quadruped

- Independently built an end-to-end sim-to-real pipeline: policy training in Isaac Lab (PhysX and Newton engines), software-in-the-loop (SIL) sim-to-sim validation in MuJoCo that runs the same C++/ONNX deployment stack over DDS, and on-board deployment on an NVIDIA Jetson; achieved real-world walking in Aug 2026.
- Traced sim-to-sim transfer failures to an IMU frame defined at the center of mass instead of the body frame; identified engine-specific friction handling, and found that Isaac Lab's Go2 model ignores the knee gear ratio (≈1.92), which I reported upstream (isaac-sim/IsaacLab#7479).
- Deployed a proprioception-only stair-climbing policy on the real Go2 by warm-starting from a TARLoco policy and fine-tuning it with additional domain randomization and compliant terrain (video: youtu.be/VR4S4SszCUg).
- Deployed a policy with a commanded force budget on the real Go2, switching from yielding to an external pull to pulling a load forward (video: youtu.be/wGCxth3PZbg).

Friction-aware quadrupedal locomotion under external forces

- Manuscript in preparation.

Lab project: end-to-end autonomous driving, sim-to-real with the MORAI simulator

- Developed visualization and analysis tools: automated evaluation using simulation metrics and derived metrics (e.g., jolt), per-segment distributions, and a multi-camera playback tool (9 views × 5 time steps).

Undergraduate Research Intern, MLML, KAIST Jun 2023 – Aug 2023
Machine Learning and Mind Lab; Advisor: Prof. Sungjin Ahn

- Ported the MuJoCo Maze environment to the codebase of Choreographer (ICLR 2023), a world-model-based skill learning method, and modified its architecture to support discrete action spaces.
- Built a visualization tool for the world model's imagined states; assisted with hyperparameter tuning and experiments.

Independent Research: Neural Fictitious Self-Play (NFSP) 2025

- Improved the sample efficiency of the NFSP baseline on Leduc Hold'em (OpenSpiel) by about 30% by replacing its MLP with a GRU and then a Transformer encoder over heterogeneous tokens (cards, position/stack, action history).

Publications

Manuscript in preparation

- **Sungwon Yang** and David Hyunchul Shim. Friction-aware quadrupedal locomotion under external forces.

Honors & Awards

3rd Place, Chief of Naval Operations Award Nov 2024
Military AI Contest (MAICON) 2024, Ministry of National Defense, Korea

- 3rd of 8 finalist teams (434 participants in the preliminary round); built an autonomous tank that follows a track and identifies friend or foe.
- Fine-tuned a YOLOv11n detector for occlusion robustness and optimized it with TensorRT to free compute for the line-following controller.

Industry Experience

Intern, Pixelro Jun 2022 – Aug 2022
Digital healthcare startup spun off from Samsung Electronics

- Proposed replacing an outsourced gaze and pose tracking module with an open-source deep learning library from Google; reviewed licensing and obtained approval, saving over KRW 20 million in planned costs.
- Planned features for a vision-test and macular degeneration screening app, and developed an eye-exercise game web app full stack.

Military Service

Republic of Korea Army Oct 2023 – Apr 2025
Microwave communications operation and maintenance specialist

Selected Projects

- **AlphaGo Zero reimplementation for Gomoku.** Implemented AlphaGo Zero from scratch in PyTorch, adapted it to Gomoku with a smaller board and network, and explored parallelization to speed up training.

Relevant Coursework

Deep Reinforcement Learning, Introduction to Artificial Intelligence, Introduction to Computer Vision, Machine Learning, Intelligent Robot Design and Programming, Dynamics, Applied Differential Equations, Applied Electronics, Systems Programming

Technical Skills

Programming: Python, C/C++, JavaScript/HTML
Robotics & Simulation: Isaac Lab, Isaac Sim, MuJoCo, Newton, Unitree Go2, NVIDIA Jetson
Machine Learning: PyTorch, RSL-RL, OpenSpiel, YOLO, TensorRT
AI-Assisted Engineering: Claude Code, Codex; delegating experiments, debugging, and code audits to coding agents, with manual verification in simulation
Languages: Korean (native), English